Signals and Systems Tutorial 5-2

Sun Yifei

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Tutorial 5

Problems: 5.24 5.36

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$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

TABLE 5.1 PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Section	Property	Aperiodic Signal	Fourier Transform
		$x[n]$ $y[n]$	$\left.\begin{array}{l}X(e^{j\omega})\\Y(e^{j\omega})\end{array}\right\}$ periodic with period 2π
5.3.2	Linearity	$ax[n]+by[n]$	$aX(e^{j\omega})+bY(e^{j\omega})$
5.3.3	Time Shifting	$x[n-n_0]$	$e^{-j\omega n_0}X(e^{j\omega})$
5.3.3	Frequency Shifting	$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$
5.3.4	Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
5.3.6	Time Reversal	$x[-n]$	$X(e^{-j\omega})$
5.3.7	Time Expansion	$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$	$X(e^{jk\omega})$
5.4	Convolution	$x[n]*y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
5.5	Multiplication	$x[n]y[n]$	$\frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
5.3.5	Differencing in Time	$x[n]-x[n-1]$	$(1-e^{-j\omega})X(e^{j\omega})$
5.3.5	Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1-e^{-j\omega}}X(e^{j\omega}) + \pi X(e^{j0})\sum_{k=-\infty}^{+\infty}\delta(\omega-2\pi k)$
5.3.8	Differentiation in Frequency	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$
5.3.4	Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \Re e\{X(e^{j\omega})\} = \Re e\{X(e^{-j\omega})\} \\ \Im m\{X(e^{j\omega})\} = -\Im m\{X(e^{-j\omega})\} \\ \lvert X(e^{j\omega})\rvert = \lvert X(e^{-j\omega})\rvert \\ \sphericalangle X(e^{j\omega}) = -\sphericalangle X(e^{-j\omega}) \end{cases}$
5.3.4	Symmetry for Real, Even Signals	$x[n]$ real an even	$X(e^{j\omega})$ real and even
5.3.4	Symmetry for Real, Odd Signals	$x[n]$ real and odd	$X(e^{j\omega})$ purely imaginary and odd
5.3.4	Even-odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}v\{x[n]\}$ $[x[n]$ real$]$ $x_o[n] = \mathcal{O}d\{x[n]\}$ $[x[n]$ real$]$	$\Re e\{X(e^{j\omega})\}$ $j\Im m\{X(e^{j\omega})\}$
5.3.9	Parseval's Relation for Aperiodic Signals	$\sum_{n=-\infty}^{+\infty}\lvert x[n]\rvert^2 = \frac{1}{2\pi}\int_{2\pi}\lvert X(e^{j\omega})\rvert^2 d\omega$	

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TABLE 5.2 BASIC DISCRETE-TIME FOURIER TRANSFORM PAIRS

Signal	Fourier Transform	Fourier Series Coefficients (if periodic)
$\sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	a_k
$e^{j\omega_0 n}$	$2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi l)$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} 1, & k = m, m \pm N, m \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\cos \omega_0 n$	$\pi \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} \frac{1}{2}, & k = \pm m, \pm m \pm N, \pm m \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\sin \omega_0 n$	$\frac{\pi}{j} \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi r}{N}$ $a_k = \begin{cases} \frac{1}{2j}, & k = r, r \pm N, r \pm 2N, \ldots \\ -\frac{1}{2j}, & k = -r, -r \pm N, -r \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$x[n] = 1$	$2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - 2\pi l)$	$a_k = \begin{cases} 1, & k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$
Periodic square wave $x[n] = \begin{cases} 1, & \|n\| \leq N_1 \\ 0, & N_1 < \|n\| \leq N/2 \end{cases}$ and $x[n + N] = x[n]$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{\sin[(2\pi k/N)(N_1 + \frac{1}{2})]}{N \sin[2\pi k/2N]}, \ k \neq 0, \pm N, \pm 2N, \ldots$ $a_k = \frac{2N_1 + 1}{N}, \ k = 0, \pm N, \pm 2N, \ldots$
$\sum_{k=-\infty}^{+\infty} \delta[n - kN]$	$\frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{1}{N}$ for all k

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$a^n u[n], \quad \lvert a\rvert < 1$	$\dfrac{1}{1 - ae^{-j\omega}}$	—
$x[n]\begin{cases} 1, & \lvert n\rvert \le N_1 \\ 0, & \lvert n\rvert > N_1 \end{cases}$	$\dfrac{\sin[\omega(N_1 + \frac{1}{2})]}{\sin(\omega/2)}$	—
$\dfrac{\sin Wn}{\pi n} = \dfrac{W}{\pi}\operatorname{sinc}\left(\dfrac{Wn}{\pi}\right)$ $0 < W < \pi$	$X(\omega) = \begin{cases} 1, & 0 \le \lvert\omega\rvert \le W \\ 0, & W < \lvert\omega\rvert \le \pi \end{cases}$ $X(\omega)$ periodic with period 2π	—
$\delta[n]$	1	—
$u[n]$	$\dfrac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{+\infty} \pi\delta(\omega - 2\pi k)$	—
$\delta[n - n_0]$	$e^{-j\omega n_0}$	—
$(n + 1)a^n u[n], \quad \lvert a\rvert < 1$	$\dfrac{1}{(1 - ae^{-j\omega})^2}$	—
$\dfrac{(n + r - 1)!}{n!(r - 1)!} a^n u[n], \quad \lvert a\rvert < 1$	$\dfrac{1}{(1 - ae^{-j\omega})^r}$	—

Problem 5.24

5.24. Determine which, if any, of the following signals have Fourier transforms that satisfy each of the following conditions:

1. $\Re e\{X(e^{j\omega})\} = 0.$
2. $\Im m\{X(e^{j\omega})\} = 0.$
3. There exists a real α such that $e^{j\alpha\omega}X(e^{j\omega})$ is real.
4. $\int_{-\pi}^{\pi} X(e^{j\omega})d\omega = 0.$
5. $X(e^{j\omega})$ periodic.
6. $X(e^{j0}) = 0.$

(a) $x[n]$ as in Figure P5.24(a)
(b) $x[n]$ as in Figure P5.24(b)
(c) $x[n] = (\frac{1}{2})^n u[n]$
(d) $x[n] = (\frac{1}{2})^{|n|}$
(e) $x[n] = \delta[n-1] + \delta[n+2]$
(f) $x[n] = \delta[n-1] + \delta[n+3]$
(g) $x[n]$ as in Figure P5.24(c)
(h) $x[n]$ as in Figure P5.24(d)
(i) $x[n] = \delta[n-1] - \delta[n+1]$

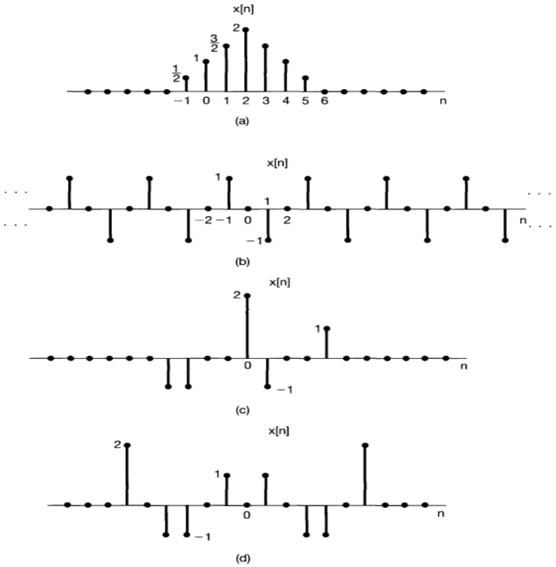

Fig P5.24

Problem 5.24

5.24. Determine which, if any, of the following signals have Fourier transforms that satisfy each of the following conditions:

1. $\Re e\{X(e^{j\omega})\} = 0.$ real & odd
2. $\Im m\{X(e^{j\omega})\} = 0.$ real & even
3. There exists a real α such that $e^{j\alpha\omega}X(e^{j\omega})$ is real. real & even after time-shifting
4. $\int_{-\pi}^{\pi} X(e^{j\omega})d\omega = 0.$ x[0]=0
5. $X(e^{j\omega})$ periodic. DT signal
6. $X(e^{j0}) = 0.$ $\sum_{n=-\infty}^{\infty} x[n] = 0$

(a) $x[n]$ as in Figure P5.24(a)
(b) $x[n]$ as in Figure P5.24(b)
(c) $x[n] = (\frac{1}{2})^n u[n]$
(d) $x[n] = (\frac{1}{2})^{|n|}$ even
(e) $x[n] = \delta[n-1] + \delta[n+2]$
(f) $x[n] = \delta[n-1] + \delta[n+3]$
(g) $x[n]$ as in Figure P5.24(c)
(h) $x[n]$ as in Figure P5.24(d)
(i) $x[n] = \delta[n-1] - \delta[n+1]$
odd

	1	2	3	4	5	6
a			√		√	
b	√		√	√	√	√
c					√	
d		√	√		√	
e			√	√	√	
f			√	√	√	
g					√	√
h		√	√	√	√	
i	√			√	√	√

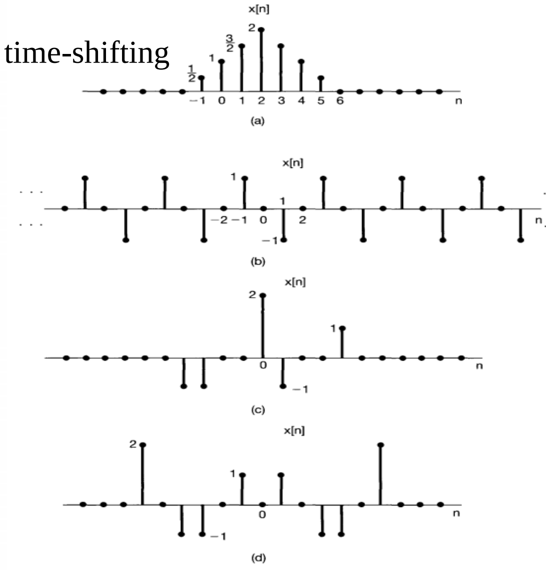

Fig P5.24

even

odd, periodic

even

Answer 5.24

5.24 (1) for $\mathrm{Re}\{X(e^{jw})\}$ to be zero, the signal must be real and odd. Only signal (b) and (i) Are real and odd.

(2) $\mathrm{Im}\{X(e^{jw})\}$ to be zero , the signal must be real and even Only signal (d) and (h) Are real and even

(3) Assume $Y(e^{jw}) = e^{jaw}X(e^{jw})$. Using the time shifting property of the Fourier transform We have y[n]=$x[n+a]$, if $Y(e^{jw})$ is real, then y[n] is real and even (assuming that x[n] is real). Therefore, x[n] has to be symmetric about a/ this is true only for signal (a) , (b), (d) , (e) , (f) , and (h).

(4) since $\int_{-\pi}^{\pi} X(e^{jw})dw = 2\pi x[0]$,the given condition is satisfied only is x[0]=0. this is True for signal (b), (e) , (f) , (h) , and (i).

(5) $X(e^{jw})$ is always periodic with period 2π. Therefore ,all signal satisfy this condition

(6) since $x(e^{j0}) = \sum_{n=-\infty}^{\infty} x[n]$,the given condition is satisfied only the samples of the signal Add up to zero. This is true for signal (b), (g) , and (i).

Problem 5.36

5.36. **(a)** Let $h[n]$ and $g[n]$ be the impulse responses of two stable discrete-time LTI systems that are inverses of each other. What is the relationship between the frequency responses of these two systems?

(b) Consider causal LTI systems described by the following difference equations. In each case, determine the impulse response of the inverse system and the difference equation that characterizes the inverse.

(i) $y[n] = x[n] - \frac{1}{4}x[n-1]$

(ii) $y[n] + \frac{1}{2}y[n-1] = x[n]$

(iii) $y[n] + \frac{1}{2}y[n-1] = x[n] - \frac{1}{4}x[n-1]$

(iv) $y[n] + \frac{5}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n] - \frac{1}{4}x[n-1] - \frac{1}{8}x[n-2]$

(v) $y[n] + \frac{5}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n] - \frac{1}{2}x[n-1]$

(vi) $y[n] + \frac{5}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n]$

(c) Consider the causal, discrete-time LTI system described by the difference equation

$$y[n] + y[n-1] + \frac{1}{4}y[n-2] = x[n-1] - \frac{1}{2}x[n-2]. \qquad \text{(P5.36–1)}$$

What is the inverse of this system? Show that the inverse is not causal. Find another causal LTI system that is an "inverse with delay" of the system described by eq. (P5.36–1). Specifically, find a causal LTI system such that the output $w[n]$ in Figure P5.36 equals $x[n-1]$.

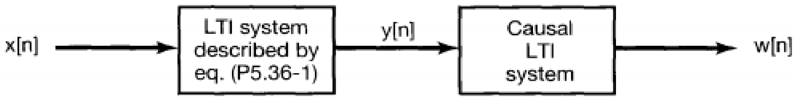

Fig P5.36

LTI by Difference Equation

- A number of DT LTI systems can be written as the following linear constant-coefficient difference equation

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

What's the frequency response?

- Taking Fourier transform on both side, we have

$$\sum_{k=0}^{N} a_k e^{-jk\omega} Y(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega} X(e^{j\omega})$$

$$\Rightarrow \qquad H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}}$$

What's the impulse response?

Answer 5.36

5.36 (a) The frequency responses are related by the following expression

$$G(e^{j\omega}) = \frac{1}{H(e^{j\omega})}$$

(b) (i) Here $H(e^{j\omega}) = 1 - \frac{1}{4}e^{-j\omega}$. Therefore, $G(e^{j\omega}) = 1 \Big/ \left(1 - \frac{1}{4}e^{-j\omega}\right)$ and $g[n] = \left(\frac{1}{4}\right)^n u[n]$ since

$$G(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{1 - \frac{1}{4}e^{-j\omega}}$$

$a^n u[n], \quad \lvert a \rvert < 1$	$\frac{1}{1 - ae^{-j\omega}}$

the difference equation relating the input $x[n]$ and output $y[n]$ is

$$y[n] - \frac{1}{4}y[n-1] = x[n]$$

(ii) Here $H(e^{j\omega}) = 1 \Big/ \left(1 + \frac{1}{2}e^{-j\omega}\right)$. Therefore, $G(e^{j\omega}) = 1 + \frac{1}{2}e^{-j\omega}$ and $g[n] = \delta[n] + \frac{1}{2}\delta[n-1]$. since

$$G(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = 1 + \frac{1}{2}e^{-j\omega}$$

$\delta[n - n_0]$	$e^{-j\omega n_0}$

the difference equation relating the input $x[n]$ and output $y[n]$ is

$$y[n] = x[n] + \frac{1}{2}x[n-1]$$

Answer 5.36

(iii) Here $H(e^{j\omega})=\left(1-\frac{1}{4}e^{-j\omega}\right)\Big/\left(1+\frac{1}{2}e^{-j\omega}\right)$. therefore , $G(e^{j\omega})=\left(1+\frac{1}{2}e^{-j\omega}\right)\Big/\left(1-\frac{1}{4}e^{-j\omega}\right)$

$$G(e^{j\omega})=\frac{1+\frac{1}{2}e^{-j\omega}}{1-\frac{1}{4}e^{-j\omega}}=\frac{(1-\frac{1}{4}e^{-j\omega})+\frac{3}{4}e^{-j\omega}}{1-\frac{1}{4}e^{-j\omega}}$$

$$G(e^{j\omega})=\frac{1+\frac{1}{2}e^{-j\omega}}{1-\frac{1}{4}e^{-j\omega}}=\frac{-2(1-\frac{1}{4}e^{-j\omega})+3}{1-\frac{1}{4}e^{-j\omega}}=-2+\frac{3}{1-\frac{1}{4}e^{-j\omega}}$$

$$g[n]=-2\delta[n]+3\left(\tfrac{1}{4}\right)^n u[n]$$

$ax[n]+by[n]$	$aX(e^{j\omega})+bY(e^{j\omega})$
$a^n u[n]$, $\|a\|<1$	$\frac{1}{1-ae^{-j\omega}}$
$\delta[n]$	1

since $$G(e^{j\omega})=\frac{Y(e^{j\omega})}{X(e^{j\omega})}=\frac{1+\frac{1}{2}e^{-j\omega}}{1-\frac{1}{4}e^{-j\omega}}$$

the difference equation relating the input $x[n]$ and output $y[n]$ is

$$y[n]-\frac{1}{4}y[n-1]=x[n]+\frac{1}{2}x[n-1]$$

Answer 5.36

(iv) Here $H(e^{j\omega}) = \left(1-\frac{1}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)\Big/\left(1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)$ .therefore

$$G(e^{jw}) = \frac{1+\frac{5}{4}e^{-jw}-\frac{1}{8}e^{-2jw}}{1-\frac{1}{4}e^{-jw}-\frac{1}{8}e^{-2jw}}$$

$$= 1 + \frac{\frac{3}{2}e^{-jw}}{1-\frac{1}{4}e^{-jw}-\frac{1}{8}e^{-2jw}} \Rightarrow \frac{\sim}{(1+xe^{-jw})(1+ye^{-jw})}, \text{ where } \begin{cases} xy = -\frac{1}{8} \\ x+y = -\frac{1}{4} \end{cases} \Rightarrow \begin{cases} x = -\frac{1}{2} \\ y = \frac{1}{4} \end{cases}$$

$$= 1 + \frac{\frac{3}{2}e^{-jw}}{(1-\frac{1}{2}e^{-jw})(1+\frac{1}{4}e^{-jw})} \Rightarrow 1 + \frac{x}{(1-\frac{1}{2}e^{-jw})} + \frac{y}{(1+\frac{1}{4}e^{-jw})}, \text{ where } \begin{cases} x+y=0 \\ \frac{1}{4}x-\frac{1}{2}y=\frac{3}{2} \end{cases} \Rightarrow \begin{cases} x=2 \\ y=-2 \end{cases}$$

$$= 1 + \frac{2}{1-\frac{1}{2}e^{-jw}} - \frac{2}{1+\frac{1}{4}e^{-jw}}$$

Answer 5.36

(iv) Here $H(e^{j\omega})=\left(1-\frac{1}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)\Big/\left(1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)$ .therefore

$$G(e^{j\omega})=\left(1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)\Big/\left(1-\frac{1}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)$$ therefore

$$G(e^{j\omega})=1+\frac{2}{1-(1/2)e^{-j\omega}}-\frac{2}{1+(1/4)e^{-j\omega}}$$

and

$$g[n]=\delta[n]+2\left(\frac{1}{2}\right)^n u[n]-2\left(-\frac{1}{4}\right)^n u[n]$$

$ax[n]+by[n]$	$aX(e^{j\omega})+bY(e^{j\omega})$
$a^n u[n],\quad \lvert a\rvert<1$	$\frac{1}{1-ae^{-j\omega}}$
$\delta[n]$	1

since

$$G(e^{j\omega})=\frac{Y(e^{j\omega})}{X(e^{j\omega})}=\frac{\left(1+\frac{5}{4}e^{-jw}-\frac{1}{8}e^{-2j\omega}\right)}{\left(1-\frac{1}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)}$$

the difference equation relating the input $x[n]$ and output $y[n]$ is

$$y[n]-\frac{1}{4}y[n-1]-\frac{1}{8}y[n-2]=x[n]+\frac{5}{4}x[n-1]-\frac{1}{8}x[n-2]$$

Answer 5.36

(v) $H(e^{j\omega}) = \dfrac{1-\frac{1}{2}e^{-j\omega}}{1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}}$.

Therefore, $G(e^{j\omega}) = \dfrac{1}{H(e^{j\omega})} = \dfrac{1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}}{1-\frac{1}{2}e^{-j\omega}} \Rightarrow \dfrac{a(1-\frac{1}{2}e^{-j\omega})^2+b(1-\frac{1}{2}e^{-j\omega})+C}{1-\frac{1}{2}e^{-j\omega}}$,

where $\begin{cases} \frac{1}{4}a = -\frac{1}{8} \\ -a-\frac{1}{2}b = \frac{5}{4} \\ a+b+C=1 \end{cases} \Rightarrow \begin{cases} a=-\frac{1}{2} \\ b=-\frac{3}{2} \\ C=3 \end{cases}$

$$= \frac{-\frac{1}{2}(1-\frac{1}{2}e^{-j\omega})^2-\frac{3}{2}(1-\frac{1}{2}e^{-j\omega})+3}{1-\frac{1}{2}e^{-j\omega}}$$

$$= -\frac{1}{2}+\frac{1}{4}e^{-j\omega}-\frac{3}{2}+\frac{3}{1-\frac{1}{2}e^{-j\omega}}$$

$$= -2+\frac{1}{4}e^{-j\omega}+\frac{3}{1-\frac{1}{2}e^{-j\omega}}$$

Then $g[n] = -2\delta[n]+\frac{1}{4}\delta[n-1]+3\left(\frac{1}{2}\right)^n u[n]$

the difference equation relating the input $x[n]$ and output $y[n]$ is

$$y[n]-\frac{1}{2}y[n-1]=x[n]+\frac{5}{4}x[n-1]-\frac{1}{8}x[n-2]$$

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Answer 5.36

(vi) Here $H(e^{j\omega}) = 1\Big/\left(1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)$ .therefore

$$G(e^{j\omega}) = \left(1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right) \quad \text{since}$$

$$G(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \left(1+\frac{5}{4}e^{-j\omega}-\frac{1}{8}e^{-2j\omega}\right)$$

We have

$$g[n] = \delta[n] + \frac{5}{4}\delta[n-1] - \frac{1}{8}\delta[n-2]$$

and the difference equation relating the input $x[n]$ and output $y[n]$ is

$$y[n] = x[n] + \frac{5}{4}x[n-1] - \frac{1}{8}x[n-2]$$

4 8

Answer 5.36

(c) $H(e^{j\omega}) = \dfrac{e^{-j\omega} - \frac{1}{2}e^{-2j\omega}}{1 + e^{-j\omega} + \frac{1}{4}e^{-2j\omega}}$

$$G(e^{j\omega}) = \frac{1}{H(e^{j\omega})} = \frac{1 + e^{-j\omega} + \frac{1}{4}e^{-2j\omega}}{e^{-j\omega}(1 - \frac{1}{2}e^{-j\omega})}$$

$$= \frac{e^{j\omega} + 1 + \frac{1}{4}e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}}$$

$$= \frac{e^{j\omega}(1 - \frac{1}{2}e^{-j\omega}) + (-\frac{1}{2})(1 - \frac{1}{2}e^{-j\omega}) + 2}{1 - \frac{1}{2}e^{-j\omega}}$$

$$= e^{j\omega} - \frac{1}{2} + \frac{2}{1 - \frac{1}{2}e^{-j\omega}}$$

$g[n] = \delta[n+1] - \frac{1}{2}\delta[n] + 2\left(\frac{1}{2}\right)^n u[n]$ which is not causal impulse response.

$g'[n] = g[n-1] = \delta[n] - \frac{1}{2}\delta[n-1] + 2\left(\frac{1}{2}\right)^n u[n-1]$

Thank you for listening